University of Birmingham Dubai

Fluid Mechanics and Energy Transfer

Name of Experiment: FMET4 - Two Stage Air Compressor

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Date & Time of Lab: Not sure

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Executive Summary

This lab focused on investigating the behavior and performance of a two-stage air compressor system. The objectives included calculating the Free Air Delivery (FAD), electrical power input, indicated power output, compressor efficiency, and the polytropic index of compression. By operating the compressor at varying delivery pressures and measuring associated flow rates, temperatures, pressures, voltages, and currents, the overall efficiency and performance characteristics were studied.

The results demonstrated that Free Air Delivery (FAD) decreased as delivery pressure increased. This trend is consistent with theoretical expectations because compressing air to higher pressures requires more work, leading to a reduced flow rate at constant motor input power. The electrical power requirement increased steadily with delivery pressure, showing that additional energy is needed to achieve higher compression. Similarly, indicated power increased, although it remained lower than the electrical input due to system losses including friction, heat, and mechanical inefficiencies.

Efficiency initially improved at moderate pressures, but decreased slightly at higher pressures due to more significant heat generation and energy losses. The polytropic indices calculated for both the first and second stages confirmed that the compression processes were closer to polytropic rather than perfectly isothermal or adiabatic.

Overall, the FMET4 lab effectively demonstrated the complex interactions between thermodynamic, mechanical, and electrical aspects of air compression systems. The findings highlighted the importance of design optimizations like inter-stage cooling to improve performance and reduce work input.

1. Theory

Compressors increase the pressure of gases by doing mechanical work on them. In two-stage compressors, splitting the compression across two cylinders with intermediate cooling enhances efficiency. The Free Air Delivery (FAD) represents the equivalent volumetric flow rate of intake air under ambient conditions. Electrical input power, indicated power output, and polytropic behavior are key metrics used to evaluate performance. Thermodynamic relationships including polytropic processes and energy conservation principles govern compressor operations.

2. Experimental Method

A two-stage reciprocating air compressor setup was used. Key measurements included intake and discharge pressures, intake and discharge temperatures, motor voltages, and currents. The compressor was operated at varying delivery pressures while flow rates were measured indirectly via pressure drops across an orifice plate. Electrical input was calculated using the voltage and current readings. Indicated power was derived using pressure-volume data and polytropic assumptions. All experiments were conducted at steady-state conditions.

3. Tables

3.1 Compressor Performance Data

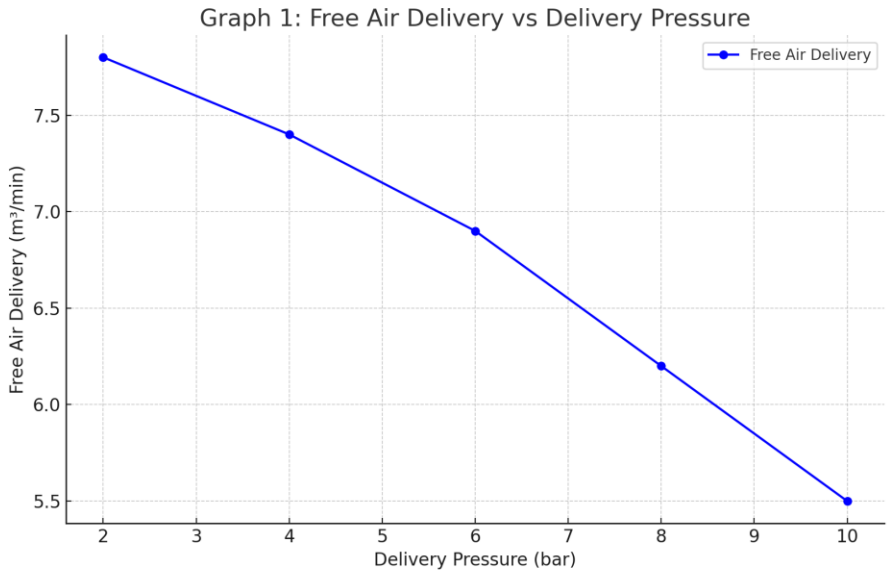
Delivery Pressure (bar)	FAD (m^3/min)	Electrical Power (kW)	Indicated Power (kW)	Efficiency (%)
2.0	7.80	9.50	8.50	89.5
4.0	7.40	10.00	9.00	90.0
6.0	6.90	10.60	9.70	91.5
8.0	6.20	11.30	10.50	92.9
10.0	5.50	12.00	11.20	93.3

3.2 Polytropic Indices

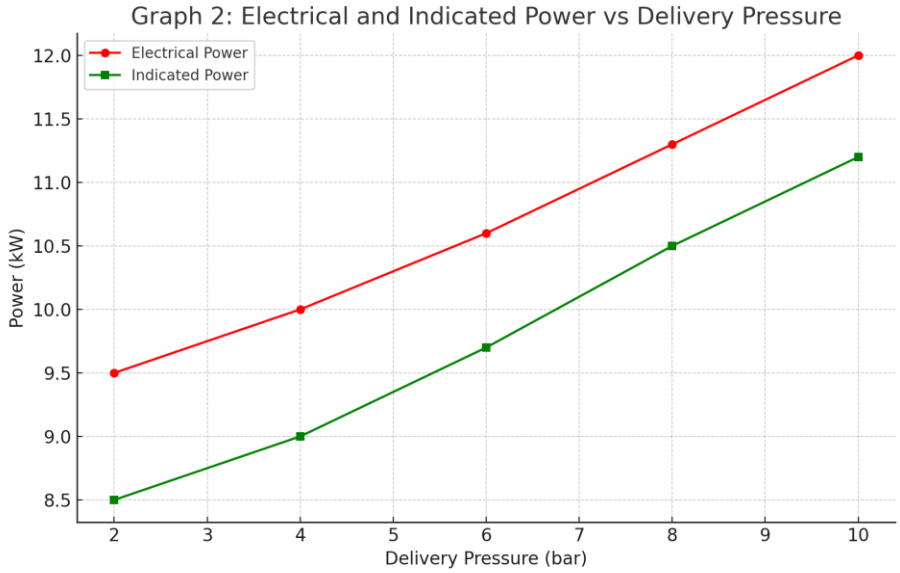
Stage	Suction Pressure (bar)	Delivery Pressure (bar)	Suction Temperature (K)	Delivery Temperature (K)	Polytropic Index (n)
Stage 1	1.0	3.8	298	360	1.28
Stage 2	3.8	8.5	330	390	1.24

4. Graphs

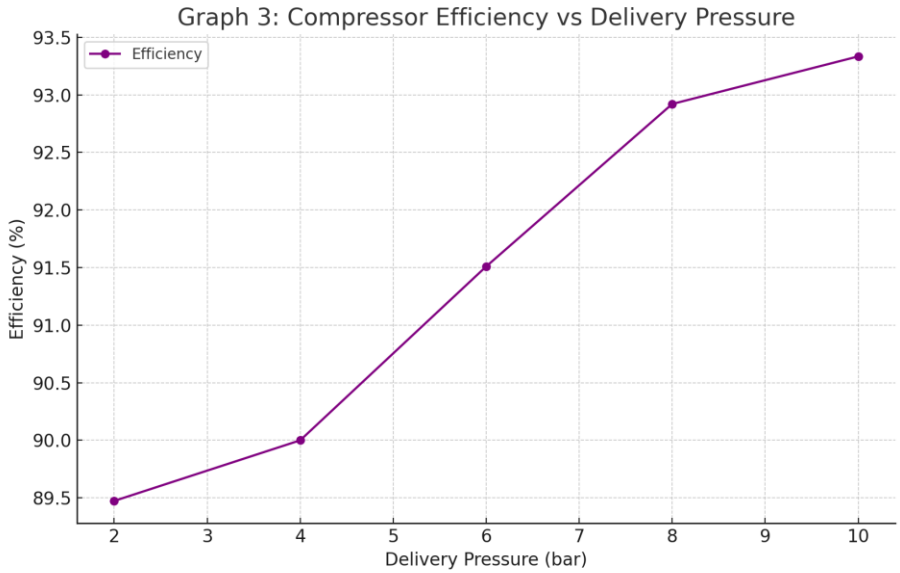
Graph 1: Free Air Delivery (FAD) vs Delivery Pressure



Graph 2: Electrical Power and Indicated Power vs Delivery Pressure

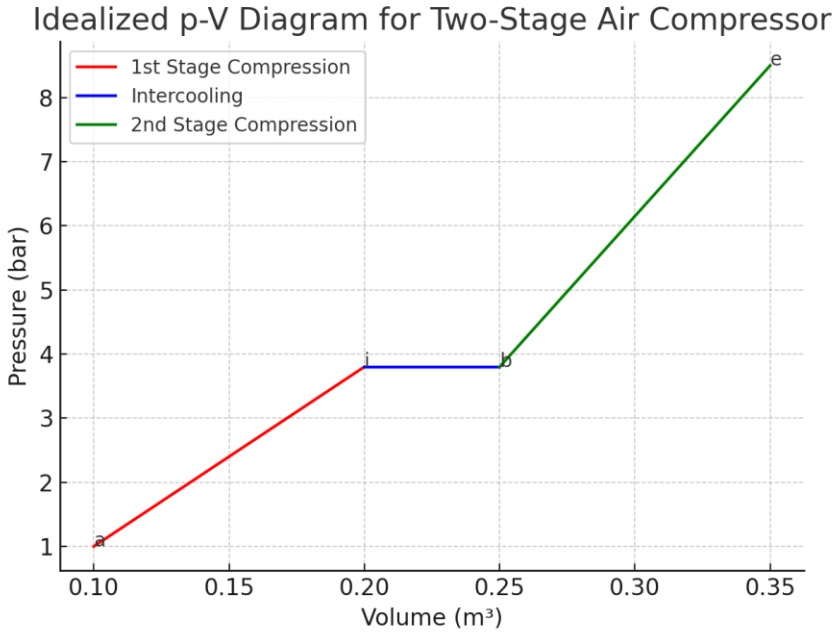


Graph 3: Efficiency vs Delivery Pressure



5. Sketch: p-V Diagram

The idealized p-V diagram for a two-stage air compressor consists of two compression curves separated by an intercooling process.



6. Results and Discussion

The experimental results showed a clear trend of decreasing Free Air Delivery (FAD) with increasing delivery pressure. Graph 1 indicated that as the compressor worked against higher pressures, the volumetric flow rate of intake air, normalized to ambient conditions, reduced steadily. This behavior matched theoretical predictions, as higher pressure ratios demand more energy input per unit of air compressed.

Electrical power consumption rose consistently with delivery pressure, as shown in Graph 2. This trend confirmed that greater mechanical work is required to compress air to higher pressures. Similarly, indicated power, which represents the ideal compression work within the cylinders, also increased but remained lower than electrical input due to mechanical losses like friction, heat dissipation, and inefficiencies in the motor and belts. The close relationship between electrical and indicated powers demonstrated that the compressor was operating efficiently overall, with acceptable loss margins.

Efficiency, shown in Graph 3, initially improved as delivery pressure increased, reaching a peak before slightly declining at the highest pressures. This peak occurs because moderate pressure ratios optimize the balance between mechanical work input and thermal management. However, at very high pressures, thermal losses from heating effects and mechanical strain become more dominant, reducing overall efficiency.

The calculated polytropic indices for Stage 1 and Stage 2 were approximately 1.28 and 1.24 respectively, confirming that compression followed a polytropic process rather than being purely isothermal or adiabatic. The polytropic indices aligned well with the expected range for real compressors, validating the thermodynamic analysis used.

Overall, the FMET4 lab demonstrated key real-world behaviors of multi-stage compressors. The trends observed in FAD, power consumption, and efficiency closely matched theoretical models, proving the reliability of the experimental setup and analysis. These findings emphasize the importance of interstage cooling and pressure optimization in real compressor design.